Is education the main driver of a high income? A CART classification tree from scratch on the Adult census data

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Code: `github.com/Petrel971/ML-Project-Bootcamp` — data: `kaggle.com/datasets/uciml/adult-census-income`

Abstract. We implement a CART classification tree from scratch in NumPy (Gini impurity, vectorised exhaustive split search, greedy recursive growth, cost-complexity pruning, impurity-based importances) and use it to ask which census variables best predict an income above $50K. On a stratified test set our tree matches `sklearn.tree.DecisionTreeClassifier` exactly in accuracy (85.7 %), F1 (0.669) and agrees on 99.98 % of predictions, while training about 20× slower. Contrary to our hypothesis, education is not the first driver: being married comes first (importance 0.43, largest ablation loss), education second (0.22). We discuss why this is association, not causation, and where a single tree fails (minority-class recall, women, variance).

1. Problem and research question

The Adult dataset (UCI, extracted from the 1994 US Census) describes 48,842 adults with 14 attributes and a binary label: does the person earn more than $50K a year? It is a classic benchmark with a moderate class imbalance (23.9 % of >50K) and a mix of numeric and categorical variables. A common belief is that *education* is what drives income. We test it:

Research question. *Is education the main driver of earning more than $50K a year, or do other factors such as marital status, age and hours worked matter more?* Starting hypothesis: education is the number-one factor.

A decision tree suits the question: it handles mixed variables without scaling, and its rules and importances are readable. The brief also requires implementing the algorithm ourselves and comparing it with a library on the same splits; CART is short enough to be written and *verified* exactly against scikit-learn.

2. Data

Variable type	Variables
Numeric (5 kept)	age, education-num (1–16, diploma level), capital-gain, capital-loss, hours-per-week
Categorical (8)	workclass, marital-status, occupation, relationship, race, sex, native-country
Dropped (2)	**fnlwgt**: census sampling weight, not a property of the person. **education**: same information as education-num; keeping both would split the importance of education in two.

Missing values only affect three categorical variables: workclass (5.7 %), occupation (5.8 %) and native-country (1.8 %). They are *informative*: only 9.4 % of the rows with a missing occupation earn >50K versus 23.9 % overall. We therefore keep them as a category "Unknown" rather than dropping or imputing them. Taken one at a time, the variables most associated with income (Cramér's V) are relationship (0.45) and marital-status (0.45), then education (0.36), occupation (0.35), age (0.32) and hours (0.27).

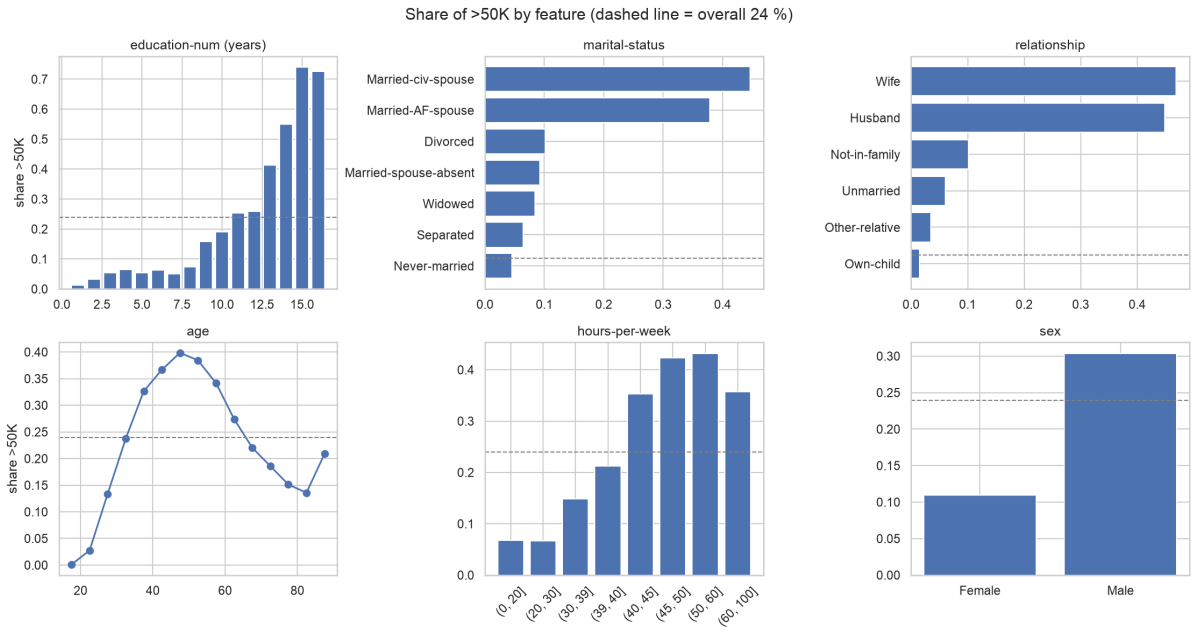


Figure 1 — Share of >50K by variable (dashed line: overall 24 %).

3. Method: CART from scratch

3.1 Model

A classification tree partitions the feature space into rectangles R_1, ..., R_M (the leaves) with rules $x_j \le s$ at internal nodes. In leaf R_m, with n_m training points, the tree returns the class proportions as probabilities and predicts the majority class:

$$\hat{p}_{mk} = \frac{1}{n_m} \sum_{x_i \in R_m} \mathbf{1}(y_i = k), \qquad \hat{y}(x) = \arg\max_k \hat{p}_{mk} \quad \text{for } x \in R_m$$

3.2 Objective

Node impurity is measured by the Gini index (entropy is also implemented):

$$G = 1 - \sum_k p_k^2 \qquad (G = 0 \text{ if pure, } G \le 0.5 \text{ with two classes})$$

A split sending n_L points left and n_R right is scored by the size-weighted impurity of its children; the best split minimises it, i.e. maximises the gain $G_{parent} - G_{split}$:

$$G_{split} = \frac{n_L}{n} G_L + \frac{n_R}{n} G_R, \qquad \Delta = G_{parent} - G_{split}$$

G is the chance that two points drawn at random in the node have different labels. Weighting by size prevents a split that isolates a single pure point from looking excellent. At the root of Adult G = 1 – (0.761² + 0.239²) = 0.364. On exercise 3B.1 of the lecture (10 clients, root G = 0.48), "income > 30?" gives G_{split} = 0.16 and "age > 40?" 0.45; our search over all thresholds finds the income split (threshold 33, same partition).

3.3 Optimisation: split search

Between two consecutive distinct values a < b of a feature, all thresholds give the same partition, so we test the midpoints (a + b)/2, as scikit-learn does. A naive search splits and scores the data for every candidate: O(d·n²). Our **vectorised** search sorts each feature once; moving to the next threshold moves one point from right to left, so the left class counts of all thresholds are the cumulative sums of the one-hot labels (right = total – left). With the points sorted by the feature, $x_{(1)} \le \dots \le x_{(n)}$:

$$N_k^L(i) = \sum_{r=1}^{i} \mathbf{1}(y_{(r)} = k), \qquad N_k^R(i) = N_k - N_k^L(i), \qquad s_i = \frac{x_{(i)} + x_{(i+1)}}{2}$$

All scores of a feature come from one NumPy expression, O(d·n log n). On continuous data it is 36× faster than the naive version at n = 250 and 260× at n = 4,000.

3.4 Construction and stopping

As in Lecture 3B, the tree is grown greedily — best question now, no look-ahead — and recursively (finding the optimal tree is NP-hard). A node becomes a leaf when max_depth is reached, it has fewer than min_samples_split or 2·min_samples_leaf points, it is pure, no valid threshold exists, or the weighted decrease $(N_t/N)(G_t - G_{split})$ is below min_impurity_decrease. Optional cost-complexity pruning (weakest link) repeatedly collapses the node with the smallest price per removed leaf, while it is at most α, where $R(t) = (N_t/N)\, G_t$ and $R(T_t)$ sums it over the leaves of the subtree:

$$g(t) = \frac{R(t) - R(T_t)}{|T_t| - 1}$$

```
fit(X, y):  root = build(X, y, depth=0)
build(X, y, depth):
    node = leaf with class proportions of y
    if stopping rule: return node
    (j, s, score) = best_split(X, y)          # vectorised, O(d n log n)
    if no split or weighted decrease too small: return node
    node.rule = (x_j <= s)
    node.left  = build(X[x_j <= s], y[...], depth + 1)
    node.right = build(X[x_j >  s], y[...], depth + 1)
    return node
```

3.5 Prediction and importance

An observation goes down the tree until a leaf (we route all rows at once with boolean masks). The importance of feature j is the weighted impurity decrease of the nodes that split on it, normalised to 1; for one-hot encoded variables we sum the importances of their columns:

$$I_j = \sum_{t\,:\ \text{split on } j} \frac{N_t}{N}\left(G_t - \frac{N_{t,L}}{N_t} G_{t,L} - \frac{N_{t,R}}{N_t} G_{t,R}\right), \qquad I_j \leftarrow \frac{I_j}{\sum_k I_k}$$

Because it is measured on the training set and favours variables with many thresholds, we also compute a **permutation importance** on the validation set: the drop in AUC when all the columns of variable v are shuffled together (π a random permutation of the rows, 10 repeats):

$$PI_v = \mathrm{AUC}(X_{val}) - \mathrm{AUC}(X_{val}^{\pi(v)})$$

3.6 Verification

Exercise 3B.1 reproduced exactly, six automated tests (`uv run pytest`) and the checkpoints of notebook 01 check: known Gini values; naive = vectorised search; the same first split as scikit-learn; a perfect fit of the training set when the depth is unlimited; identical predictions and importances for depths 1–4; the same number of leaves after pruning. On Adult, predictions agree with scikit-learn at 100 % for depths 1–4 and 99.99 % at depths 5–6; the rare differences are **ties** (two one-hot columns with exactly the same Gini), broken by column order for us and by a random feature order in scikit-learn.

4. Experimental protocol

Step	Choice	Reason
Split	Stratified 60/20/20, seed 42	Train to learn, validation to tune, test used **once**; 23.9 % >50K in each set
Encoding	One-hot, columns fixed on train	Split first, then encode: no information from validation/test (no leakage)
Scaling	None	A tree compares a variable to a threshold: scale-invariant
Tuning	max_depth 1–20, min_samples_leaf 1–100	Selected by validation F1, confirmed by 5-fold CV on train
Metrics	F1, precision, recall, ROC AUC, accuracy	A constant "≤50K" model already gets 76 % accuracy with F1 = 0
Comparison	Same splits, same hyperparameters	Ours vs DecisionTreeClassifier; also fit time and agreement rate

Train/validation/test sizes: 29,304 / 9,769 / 9,769 rows, 91 columns after encoding.

5. Results

5.1 Choosing the complexity

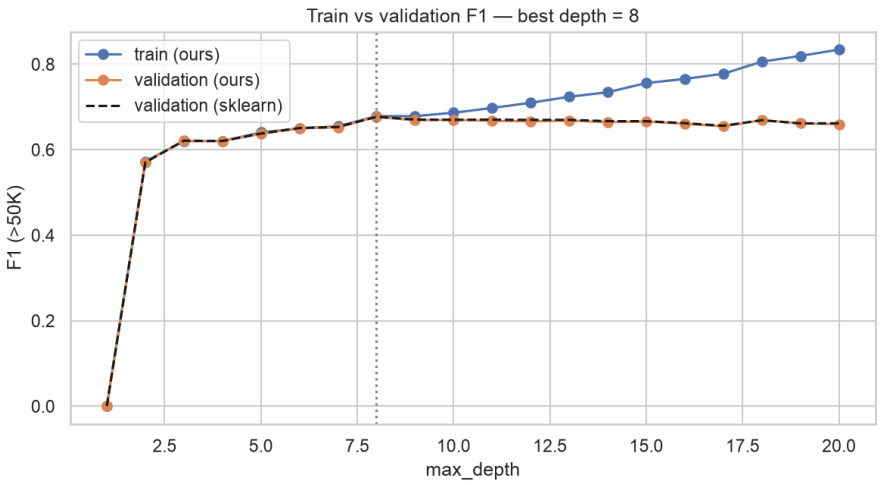


Figure 2 — Train and validation F1 by depth; our tree and scikit-learn overlap.

Validation F1 rises until depth 8 (0.677) and then stalls while train F1 keeps rising towards 0.8: deeper trees overfit (variance); shallower ones underfit (bias). Depth 1 has F1 = 0 although its AUC is 0.76: both children of the "married" split still have a ≤50K majority. min_samples_leaf = 5 slightly improves validation F1 (0.679) and AUC (0.910). **Chosen: max_depth = 8, min_samples_leaf = 5.** A 5-fold cross-validation on the training set confirms it: mean F1 reaches 0.658 ± 0.015 at depth 8, and deeper trees stay within one standard deviation (best 0.662 at depth 14); by the one-standard-error rule the simplest, depth 8, is kept.

5.2 Comparison on the test set

Model	Accuracy	Precision	Recall	F1	ROC AUC	Fit time
CART from scratch	0.857	0.751	0.602	0.669	0.906	≈ 1.0 s
scikit-learn	0.857	0.751	0.602	0.669	0.906	≈ 0.04 s
Always ≤50K	0.761	0	0	0	0.500	—

Both trees give the same prediction on 99.98 % of test rows (2 rows differ, caused by a tie) and the same metrics; the confusion matrices are identical (Figure 3). Our training is about 20× slower at every sample size from 2,000 to 29,304 rows; both grow roughly linearly in n. The gap is a constant factor: scikit-learn's splitter is compiled Cython working in place on sorted indices, while our Python recursion re-sorts features and copies sub-arrays at each node.

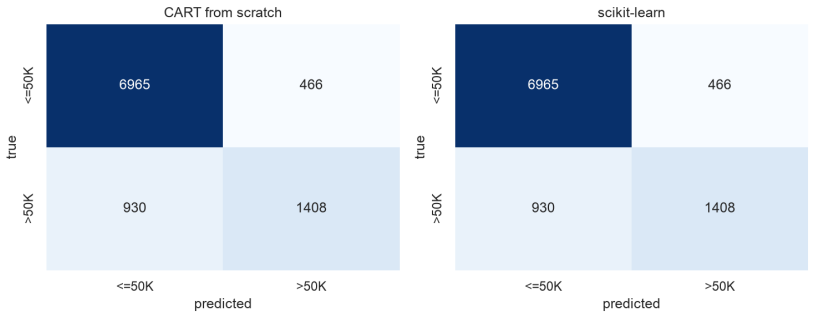


Figure 3 — Test confusion matrices (threshold 0.5).

6. Answer to the research question

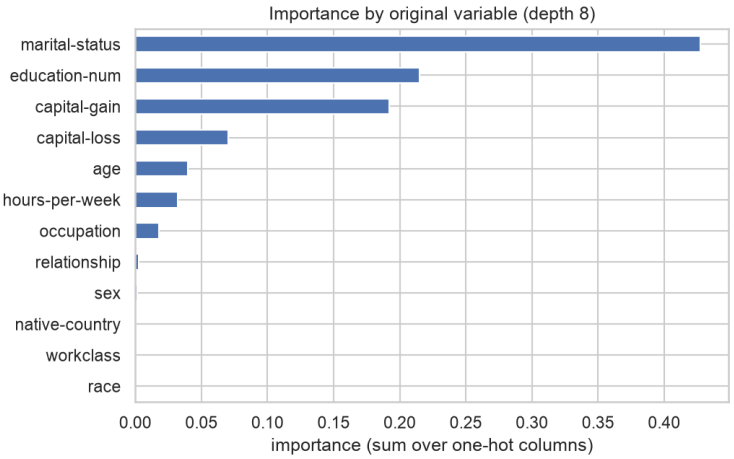


Figure 4 — Importance by original variable (depth-8 tree, one-hot columns summed).

Evidence	Family situation	Education	Others
Association alone (Cramér's V)	**0.45**	0.36	occupation 0.35, age 0.32, hours 0.27
First split	**married (civ-spouse)?**	—	—
Importance (depth 8)	**0.43**	0.22	capital-gain 0.19, capital-loss 0.07, age 0.04, hours 0.03
Permutation: validation AUC drop	**0.142**	0.048	capital-gain 0.050, age 0.032, hours 0.017
Ablation: validation AUC loss	**0.047**	0.013	capital 0.031, hours 0.007, age 0.005
Ablation: validation F1 loss	**0.122**	0.058	hours 0.048, capital 0.042, age ≈ 0

5 random splits	first split 5/5	2nd each time	—

The hypothesis is rejected. The family situation — being married and living with a spouse — is the strongest predictor by every measure: first split of the tree, twice the importance of education, and the largest loss when it is removed or shuffled — about three times education on permutation importance. Education is the **second** factor, but not by much: it is clearly second on impurity importance and on the ablation, and tied with capital gains on permutation importance (0.048 vs 0.050). Age and hours worked add little once marital status and education are known. The readable depth-3 tree tells the same story: unmarried people without capital gains have a 2–15 % chance of >50K depending on education; married people with at least an associate degree (education-num ≥ 12) reach 65 %.

Two subtleties. (1) relationship and marital-status carry the same information (Husband/Wife ⇔ married): the tree uses one and ignores the other, so removing either alone costs little; removing both costs the most. (2) Age has low importance not because it is irrelevant (V = 0.32) but because it is absorbed by marital status: importance measures what a variable adds *given the others*.

Association, not causation. Marriage does not cause a high income. In this 1994 sample, marital status is a proxy for age and career stage, for sex (married earners are mostly "Husband"), and possibly for household arrangements; capital-gain is closer to a consequence of wealth than a cause. Our conclusion is predictive: among these census variables, family situation is the best single predictor of a >50K income, and education the second.

7. Discussion: limits and when the model fails

All analyses below use the validation set or cross-validation; the test set is not used after section 5.2.
Minority class. At the default threshold the tree misses 40 % of high earners (test recall 0.60). A threshold of 0.36, chosen by 5-fold cross-validation on the training set, raises validation recall from 0.61 to 0.82 and F1 from 0.679 to 0.694, at the cost of precision (0.76 → 0.60) — the right threshold depends on the cost of each error.

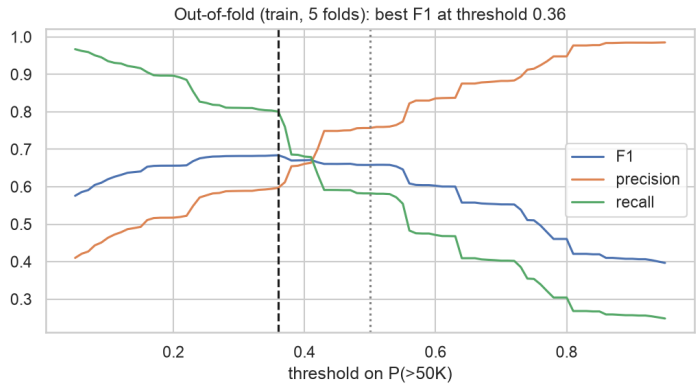


Figure 5 — Precision, recall and F1 by decision threshold (out-of-fold, 5-fold CV on train).

Errors by sex (validation). Recall is 0.54 for women against 0.63 for men (F1 0.65 vs 0.68). Only 11 % of women earn >50K, and the model predicts 7.4 %: it learns and slightly amplifies the 1994 income gap. Any use of such a model for decisions about people would need a fairness analysis.

Variance. Over 5 random splits, the first split and the top importances are unchanged (marital 0.41–0.42, education 0.22–0.23), so our answer is robust; validation F1 varies from 0.638 to 0.659 and the deep, small leaves change. **Structure.** Axis-aligned cuts approximate smooth effects by staircases; greedy growth misses interactions invisible at the first step; impurity importance is computed on the training set and favours variables with many thresholds, which is why we added ablation. **Cost.** Our implementation is about 20× slower than scikit-learn for identical results.

Improvements. Ensembles reduce the weaknesses of one tree. For discussion we ran scikit-learn's random forest (300 trees: validation F1 0.681, AUC 0.920, lower variance) and gradient boosting (F1

0.719, AUC 0.931, lower bias) on the same splits, against 0.679 / 0.910 for our tree. Both reuse the split search we implemented as their building block; a natural extension is to build a random forest on our CART.

8. Conclusion

We implemented CART from scratch and verified it against scikit-learn: same splits, same predictions on 99.98 % of test rows, same metrics (accuracy 85.7 %, F1 0.669, AUC 0.906), with a 20× slower training. Used to answer our question, it shows that **education is not the main driver of a high income in the 1994 census: family situation comes first, education second**, a predictive association that reflects age, sex and household structure rather than a causal effect.

Reproducibility

`uv sync` installs the exact versions of `uv.lock` (Python ≥ 3.11, NumPy, pandas, scikit-learn, Matplotlib, Seaborn). `uv run pytest` runs the tests; then run `notebooks/01_cart_from_scratch.ipynb` and `notebooks/02_experiments.ipynb` top to bottom (the data is downloaded from OpenML on first run and cached in `data/`, not redistributed). Seed 42 everywhere; the model code is in `src/cart.py` (NumPy only), scikit-learn is used only for splitting, metrics and the comparison.

References

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